

Mahavir Polytechnic, Nashik
Department of Computer Engineering

Year: TY

Subject: STE (315323)

UNIT 6: Software Quality Assurance

Marks: 8

COURSE LEVEL LEARNING OUTCOME:

CO6	Use quality assurance principles in software development
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THEORY LEARNING OUTCOMES (TLO'S) ALIGNED TO COS:

TLO 6.1	Differentiate between Software Quality Management and Software Quality Assurance.
TLO 6.2	Apply the phases of Software Quality Assurance in software development project
TLO 6.3	Apply software quality evaluation standards.

LEARNING CONTENT MAPPED WITH THEORY LEARNING OUTCOMES (TLO'S) AND COS:

TLO 6.1	Software Quality Management vs. Software Quality Assurance.
TLO 6.2	Phases of Software Quality Assurance: Planning, activities, audit, and review.
TLO 6.3	Quality Evaluation standards: Six Sigma, CMMI: Levels, Process areas

6.1 Software Quality Management vs. Software Quality Assurance.

- Software Quality Management (SQM) is the comprehensive system for ensuring quality.
- Software Quality Assurance (SQA) is a proactive component within SQM focused on preventing defects by establishing and monitoring processes to meet quality standards throughout the software development lifecycle.

- SQM has a broader scope, addressing the overall strategy, metrics, and improvement of quality.
- SQA is more specific, focusing on verifying that the quality system's requirements are being followed.

➤ **Software Quality Management (SQM)**

- **Scope:**

The overarching framework that manages all activities to achieve quality objectives.

- **Focus:**

Strategic planning, setting quality standards and metrics, continuous process improvement, risk management, and overall quality culture.

- **Nature:**

A proactive, forward-looking approach to "do things right".

- **Activities:**

Involves defining the entire quality system, tracking progress, analyzing processes to remove potential variations, and ensuring customer satisfaction.

➤ **Software Quality Assurance (SQA)**

- **Scope:**

A subset of SQM, focusing on the processes and activities used during software development.

- **Focus:**

Preventing defects by ensuring adherence to quality standards and procedures, providing confidence that the software will meet its requirements.

- **Nature:**

A process-oriented and proactive function that aims to prevent quality issues from occurring.

- **Activities:**

Includes activities like personnel training, process audits, documentation, setting up standard operating procedures (SOPs), and monitoring development activities.

Question: Differentiate between Software Quality Management and Software Quality Assurance.

Ans 1:

Feature	Software Quality Management (SQM)	Software Quality Assurance (SQA)
Scope	Broad; the entire quality system for the organization or project.	Narrower; a specific process within the SQM framework.

Focus	Strategic planning, setting standards, and overall quality goals.	Process adherence, preventing defects, and verifying standards are followed.
Nature	The "overall system" and strategic direction for quality.	A "technique" or system to assure quality through planned activities.
Goal	Achieve high-quality standards across the entire organization.	Ensure processes are executed correctly to prevent defects and build quality in.
Timeframe	Covers all stages of project and product delivery.	Operates throughout the entire Software Development Life Cycle (SDLC).

What is Quality Assurance (QA)?

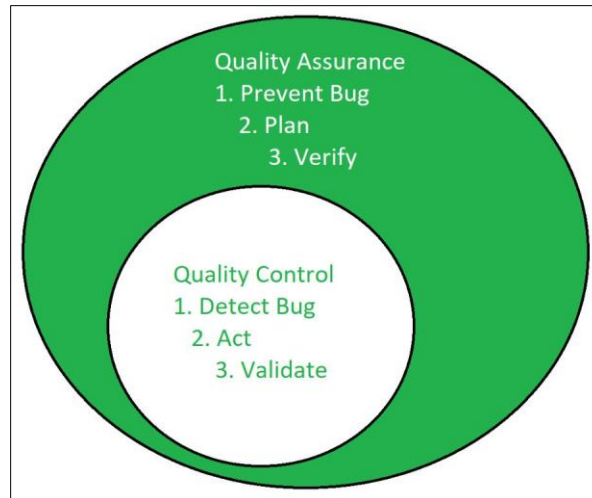
Quality assurance is a method of making the software application with fewer defects and mistakes when it is finally released to the end users. Quality Assurance is defined as an activity that ensures the approaches, techniques, methods, and processes designed for the projects are implemented correctly. It recognizes defects in the process. Quality Assurance is completed before Quality Control.

- It focuses on preventing defects.
- It is a proactive process and is preventive in nature.
- It helps to recognize flaws in the process.
- These activities monitor and verify that the processes used to manage and create deliverables have been followed.

What is Quality Control?

Quality Control is a software engineering process that is used to ensure that the approaches, techniques, methods, and processes designed for the project are followed correctly. Quality control activities operate and verify that the application meet the defined quality standards.

- It focuses on an examination of the quality of the end products and the final outcome rather than focusing on the processes used to create a product.
- It is a reactive process and is detection in nature.
- These activities monitor and verify that the project deliverables meet the defined quality standards.



➤ **Differentiate between Software Quality Assurance and Software Quality Control.**

Parameters	Quality Assurance (QA)	Quality Control (QC)
Objective	QA focuses on providing assurance that the quality requested will be achieved.	QC focuses on fulfilling the quality requested.
Technique	QA is the technique of managing quality.	QC is the technique to verify quality.
Involved in which phase?	QA is involved during the development phase.	QC is not included during the development phase.
Program execution is included?	QA does not include the execution of the program.	QC always includes the execution of the program.
Type of tool	QA is a managerial tool.	QC is a corrective tool.
Process/ Product-oriented	QA is process oriented.	QC is product oriented.
Aim	The aim of quality assurance is to prevent defects.	The aim of quality control is to identify and improve the defects.

Parameters	Quality Assurance (QA)	Quality Control (QC)
Order of execution	Quality Assurance is performed before Quality Control.	Quality Control is performed after the Quality Assurance activity is done.
Technique type	QA is a preventive technique.	QC is a corrective technique.
Measure type	QA is a proactive measure.	QC is a reactive measure.
SDLC/ STLC?	QA is responsible for the entire <u>software development life cycle</u> .	QC is responsible for the <u>software testing life cycle</u> .
Activity level	QA is a low-level activity that identifies an error and mistakes that QC cannot.	QC is a high-level activity that identifies an error that QA cannot.
Focus	Pays main focus is on the intermediate process.	Its primary focus is on final products.
Team	All team members of the project are involved.	Generally, the testing team of the project is involved.
Time consumption	QA is a less time-consuming activity.	QC is a more time-consuming activity.
Which statistical technique was applied?	<u>Statistical Process Control (SPC)</u> statistical technique is applied to Quality Assurance.	Statistical Quality Control (SQC) statistical technique is applied to Quality Control.
Example	<u>Verification</u>	Validation

6.2 Phases of Software Quality Assurance:

Software Quality Assurance (SQA) encompasses planning the overall quality strategy, actively performing SQA activities like reviews and testing, performing audits to verify process adherence, and conducting reviews to evaluate results and ensure alignment with standards. The process is iterative and continuous, typically following a **Plan-Do-Check-Act** cycle to ensure software quality throughout the development lifecycle and maintain customer satisfaction.

1. Planning

- **Define Objectives:**

Establish clear goals and objectives for software quality, aligning them with business goals and customer expectations.

- **Create the SQA Plan:**

Develop a comprehensive plan that outlines the processes, metrics, and resources needed to achieve the defined quality goals.

- **Set Success Metrics:**

Determine key performance indicators (KPIs) such as defect density, test coverage, and maintenance costs to measure quality.

- **Establish Standards:**

Identify and adopt relevant industry standards, processes, and regulatory requirements to ensure compliance.

2. Activities (Do)

- **Requirement Gathering:**

Participate in the gathering and understanding of both functional and non-functional software requirements.

- **Process Support:**

Support the development team by providing input into the software process description and adhering to defined processes.

- **Test Strategy & Execution:**

Develop and implement a multi-testing strategy, including functional, regression, integration, and performance tests, and execute them to find defects.

- **Defect Tracking:**

Track and report any defects found during testing to the senior management for resolution.

3. Audits (Check)

- **Verify Process Adherence:**

Conduct audits to verify that the development team is following the defined software processes and standards.

- **Work Product Inspection:**

Audit software work products, such as design documents, to ensure they comply with the defined standards and initial assumptions.

- **Document Compliance:**

Record any non-compliance identified during the audit and report it to the management for corrective action.

4. Reviews (Act)

- **Formal Technical Reviews:**

Conduct formal reviews of software components and designs to identify potential issues early in the development cycle.

- **Evaluate Results:**

Evaluate the outcomes of the implemented plan, checking if the set goals and objectives have been reached.

- **Continuous Improvement:**

Make any necessary changes based on the findings from the check and audit phases to refine the process and achieve better results.

6.3 Quality Evaluation standards:

Software quality evaluation uses ISO standards, primarily the ISO/IEC 25000 series (SQuaRE), to define characteristics like functionality, reliability, and usability, along with specific models like ISO/IEC 25010. These standards provide a structured approach to measuring and improving software quality, ensuring that software products meet established benchmarks and user needs. Key standards include ISO 9000 for quality management, ISO/IEC 9126 (now superseded by ISO/IEC 25010), and ISO 5055 for internal quality attributes.

➤ **Different types of ISO standards :**

Here, you will see different types of ISO standards as follows.

- **ISO 9000: 2000 -**

ISO 9000: 2000: contains Quality management systems, fundamentals, and vocabulary.

- **ISO 9000-1: 1994 -**

This series of standards includes Quality management systems and Quality assurance standards. It also includes some guidelines for selection and use.

- **ISO 9000-2: 1997 -**

- This series of standards also includes Quality management systems and Quality assurance standards. It also includes some guidelines for the application of ISO 9001, ISO 9002, and ISO 9003.

- **ISO 9000-3: 1997 -**

This series contains Quality management systems, Quality assurance standards and also includes

guidelines for the application of ISO 9001 to 1994 to the development, supply, installation, and maintenance of computer installation.

- **ISO 9001: 1994 -**
This series of standards has Quality systems and a Quality assurance model. This model helps in design, development, production, installation, and service.
- **ISO 9001: 2000 -**
This series of standards also includes Quality management systems.
- **ISO 9002: 1994 -**
This series of standards also includes some Quality systems. This Quality assurance model used in production, installation, and servicing.
- **ISO 9003: 1994 -**
This series of standards also includes some Quality systems. This Quality assurance model used in the final inspection and test.
- **ISO 9004: 2000 -**
This series of standards include some Quality management systems. It also includes some guidelines for performance improvements.
- **ISO 9039: 1994 -**
This series of standards include some Optics and Optical Instruments. It includes quality evaluation of optical systems and determination of distortion.
- **ISO/IEC 9126-1: 2001 -**
This series of standards has information technology. It also includes some software products, quality models.
- **ISO/IEC 9040: 1997 -**
This series of standards has information technology. It also includes open system interconnection and Virtual terminal basic class service.
- **ISO/IEC 9041-1: 1997 -**
This series of standards has information technology. It also includes open system interconnection, Virtual terminal basic class service protocol, and specification.
- **ISO/IEC 9041-2: 1997 -**
This series of standards include information technology, open system interconnection, Virtual terminal basic class protocol, and Protocol implementation conformance statement (PICS) proforma.
- **ISO/IEC 9075-9: 2001 -**
This series of standards has information technology, Database languages, and SQL/MED(Management of External Data).

- **ISO/IEC 9075-10: 2000 -**
This series of standards has information technology, Database languages, and SQL/OLB (Object Language Bindings).
- **ISO/IEC 9075-13: 2002 -**
This series of standards has information technology, Database languages, SQL routines, and Java Programming language. (SQL/JRT).

➤ **How These Standards Are Used**

- **Structured Assessment:**

They provide a framework for developers and evaluators to measure and assess various aspects of software quality.

- **Process Improvement:**

Standards like ISO 9001 help establish consistent processes that can lead to improved product quality over time.

- **Benchmarking:**

They allow organizations to benchmark their software quality against industry best practices.

6.3.1 Six Sigma:

- Six Sigma is a methodology that helps organizations in making their process better and more efficient by identifying and removing errors and variations.
- Variations in processes can lead to errors, these errors can lead to product defects and product defects can lead to poor customer satisfaction.
- By reducing variation and errors Six Sigma can reduce process costs and increase customer satisfaction.
- Six Sigma was introduced in 1986 by an American Engineer Bill Smith. He introduced this term while working at Motorola.
- Industries like manufacturing, service industry, government agencies, aerospace, and e-commerce use Six Sigma to improve their processes and product quality.

➤ **Methodologies & Frameworks**

- **DMAIC (Define, Measure, Analyze, Improve, Control):** Used for improving existing software development processes.
- **DMADV (Define, Measure, Analyze, Design, Verify):** Ideal for designing new software systems or processes to meet Six Sigma standards from the beginning.

6.3.2 CMMI: Levels:

The five CMMI (Capability Maturity Model Integration) levels in software engineering represent an organization's journey toward process excellence, from Level 1 (Initial) where processes are ad hoc, to Level 5 (Optimizing), focused on continuous improvement and innovation. The levels are: Initial, Managed, Defined, Quantitatively Managed, and Optimizing.

➤ **Here is a breakdown of each CMMI maturity level:**

- **Level 0: Incomplete**
 - Describes processes that are either non-existent or incomplete, leaving work unfinished and unaddressed.
- **Level 1: Initial**
 - Processes are characterized by their unpredictability and often reactive nature, with little to no formal process in place.
 - Work is completed, but it often fails to meet project timelines and budgets.
- **Level 2: Managed**
 - Processes are planned, performed, measured, and controlled at the project level, not yet on a company-wide basis.
 - This level focuses on establishing basic project management policies and leveraging past project successes.
- **Level 3: Defined**
 - Processes are well-defined, documented, and standardized across the entire organization.
 - Organizations at this level proactively understand and document their business goals and can address them through tailored standards and procedures.
- **Level 4: Quantitatively Managed**
 - Processes are measured and controlled using statistical and quantitative techniques.
 - This data-driven approach ensures process performance is predictable and aligned with stakeholder needs.
- **Level 5: Optimizing**
 - The organization focuses on continuous improvement through both incremental enhancements and innovative advancements.
 - Processes are agile and stable, allowing the organization to adapt to changing environments and pursue new opportunities.

6.3.3 Process areas: Key Process Areas (KPA)

Each of these KPA (Key Process Areas) defines the basic requirements that should be met by a software process to satisfy the KPA and achieve that level of maturity.

Conceptually, key process areas form the basis for management control of the software project and establish a context in which technical methods are applied, work products like models, documents, data, reports, etc. are produced, milestones are established, quality is ensured and change is properly managed.

Important Questions

1. Differentiate between Software Quality Management and Software Quality Assurance.
2. Apply the phases of Software Quality Assurance in software development project.
3. Describe six sigma. State operations in DMADV/DMAIC.
4. Describe CMMI. Give significance of each level